Year 12 Further Maths - Vectors

Similar questions - Modelling and geometry in three dimensions

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The points $P(2,6,-10)$ and $Q(6,2,-4)$ are on the ceiling of a long straight underground tunnel. A ventilation shaft must be dug from the point M on the ceiling midway between P and Q to horizontal ground level (where the z-coordinate is 0). The shaft must be perpendicular to the tunnel. The path of the shaft is modelled by $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ where $\mathbf{a}$ is the position vector of M and $\mathbf{b} = \begin{pmatrix} 1 \\ s \\ t \end{pmatrix}$.

(a) Show that $s = 1 + 1.5t$. **[3]**

(b) Show that at the point where the shaft reaches the ground $\lambda = \frac{c}{t}$, where c is a constant to be determined. **[3]**

(c) Using parts (a) and (b), determine the shortest possible length of the shaft, in exact form. **[6]**

(d) Explain what $\mathbf{b} \times \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \neq \mathbf{0}$ means about the direction of the shaft. **[1]**

2 The line segment AB is a diameter of a sphere S. The point C is **any** point on the surface of S.

(a) Explain why $\overrightarrow{AC} \cdot \overrightarrow{BC} = 0$ for all positions of C. **[3]**

(b) A is the point $(3,4,-2)$ and B is $(7,0,4)$. Given that C has coordinates $(2p, p, 1)$, determine the possible positions of C. **[6]**

Question 1

(a) $M = \left(\frac{2+6}{2}, \frac{6+2}{2}, \frac{-10-4}{2}\right) = (4,4,-7)$. $\overrightarrow{PQ} = \begin{pmatrix} 4 \\ -4 \\ 6 \end{pmatrix}$ and the shaft is perpendicular to the tunnel: $4-4s+6t=0$, $s = 1+1.5t$

(b) $z = -7+\lambda t = 0 \Rightarrow \lambda = \dfrac{7}{t}$, so $c = 7$

(c) Length $= |\lambda||\mathbf{b}| = \dfrac{7}{|t|}\sqrt{1+(1+1.5t)^2+t^2}$. Squaring and dividing by 49: $\dfrac{2+3t+3.25t^2}{t^2} = 3.25+\dfrac{3}{t}+\dfrac{2}{t^2}$. With $u = \frac{1}{t}$: $2u^2+3u+3.25$ is minimised at $u = -\frac{3}{4}$, value $3.25-\frac{9}{8} = \frac{17}{8}$. Shortest length $= 7\sqrt{\frac{17}{8}} = \dfrac{7\sqrt{34}}{4} = 10.2$ (when $t = -\frac{4}{3}$, $s = -1$)

(d) The shaft is not vertical (not parallel to the z-axis)

Question 2

(a) The angle in a semicircle is a right angle, so $AC \perp BC$ and the scalar product is zero

(b) $\overrightarrow{AC} = \begin{pmatrix} 2p-3 \\ p-4 \\ 3 \end{pmatrix}$, $\overrightarrow{BC} = \begin{pmatrix} 2p-7 \\ p \\ -3 \end{pmatrix}$. Dot product: $(2p-3)(2p-7)+p(p-4)-9 = 5p^2-24p+12 = 0$, $p = \dfrac{12 \pm 2\sqrt{21}}{5}$. $C = \left(\frac{24\pm4\sqrt{21}}{5}, \frac{12\pm2\sqrt{21}}{5}, 1\right)$, i.e. $(8.47, 4.23, 1)$ or $(1.13, 0.567, 1)$